XML

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Agenda

- XML in General
- XML syntax
- Validation
- XML Namespaces
- DTD

GenCode → SGML

- GenCode (1967)
  - One universal language
- GML
  - IBM
- SGML (1986)

```
<!DOCTYPE motd [ <!E
<motd>
<!-- created: 2003-12-12-->
 <sentence>Do not throw
 out the <keep>baby</>
 with the
 <refuse>dirty</>,
 <refuse>stinky</>,
 <refuse>bathwater</>.
 </>
 <!-- finish this later-->
</motd>
```

SGML

SGML

- ISO 8879
- Standard Generalized Markup Language
- How to specify a document markup language or a tag set
- Language for defining markup languages

SGML → XML

- **XML is an application of SGML**
- Simplified
- Subset
- More accessible
- Stricter syntax
- Less flexibility, faster parsing

XML - eXtensible Markup Language

- World Wide Web Consorcium (W3C) Recommendation (www.w3.org)
- Extensible Markup Language (XML) 1.0 (Fifth Edition)
- *"Its goal is to enable generic SGML to be served, received, and processed."*

XML - Extensible Markup Language

- It is only a language
- Metalanguage - for creating new languages
- Data carrier – not for displaying data
- Rules for syntax, tags composition, etc.
- The behavior of XML processor ↓

XML Processor

- Software (app, module) which reads XML documents
- Provides access to the XML content (and structure)
- XML specification describes the **required behavior** of an XML processor

XML Versions

- 1.0, 1.1
- Only small differencies
- 1.1
  - Forward compatible
  - Processor recognizes characters at the end of line
  - Character normalization and control

Use of XML

- **Semi-structured** data
- Data presentation made by a different resource
- Data sharing
  - Single file
  - Multiple files
    - Like database
- Easy to change software, carrying medium

Usage Examples

- Configurations, settings, data exchange
- Sitemaps
- Data exports
- Aggregation services in general
- Web services, in general

Usage Examples

- Android UI
- IIS configuration (Windows web server)
- NetBeans, PHP Storm (various settings)
- E-shop and price comparators
  - e.g., Heureka.cz, zbozi.cz

```xml
<?xml version="1.0" encoding="utf-8"?>
<android.support.v4.widget.DrawerLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/drawer_layout"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="@android:color/background_light">

    <FrameLayout xmlns:android="http://schemas.android.com/apk/res/android"
        android:id="@+id/oneFragment"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:padding="0dp">

    </FrameLayout>

    <LinearLayout
        android:id="@+id/left_drawer"
        android:layout_width="260dp"
        android:layout_height="fill_parent"
        android:layout_gravity="start"
        android:background="#F5F3F0"
        android:orientation="vertical">

        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:background="#cccccc"
            android:orientation="vertical">

            <Button
```

Heureka (price comparison) Products Feed


```
<IMGURL_ALTERNATIVE>http://obchod.cz/mobily/nokia-5800-xpressmusic/obrazek2.jpg</IMGURL_ALTERNATIVE>
<VIDEO_URL>http://www.youtube.com/watch?v=KjR759oWF7w</VIDEO_URL>
<PRICE_VAT>6000</PRICE_VAT>
<HEUREKA_CPC>5,8</HEUREKA_CPC>
<MANUFACTURER>NOKIA</MANUFACTURER>
<CATEGORYTEXT>Elektronika | Mobilní telefony</CATEGORYTEXT>
<EAN>6417182041488</EAN>
<PRODUCTNO>RM-559394</PRODUCTNO>
<PARAM>
  <PARAM_NAME>Barva</PARAM_NAME>
  <VAL>černá</VAL>
</PARAM>
<DELIVERY_DATE>2</DELIVERY_DATE>
<DELIVERY>
  <DELIVERY_ID>CESKA_POSTA</DELIVERY_ID>
  <DELIVERY_PRICE>120</DELIVERY_PRICE>
```

Another Examples

- GPS eXchange Format (GPX)
- Information systems
  - Exchange data with other applications
- Various services
  - openweathermap.org
  - RSS
- www.w3schools.com/xml/xml_examples.asp

Even More Examples

- MathML – mathematical (and other) formulas
- Office Open XML – MS Office .docx, .xlsx,…
- ODF –OpenOffice, LibreOffice
- SVG – vector graphics
- SCORM – e-learning courses
- UIML – GUI developement
- And many others

XML based languages

SGML

XML

SVG, DocBook, …

Related to XML

- DTD (comes from SGML)
- XML Schema, RelaxNG,…
- XSL – transformation and formatting
- XPath – reference in the document
- XLink, XPointer, XQuery
- …and more

The Influence of XML to HTML

- Reformulation of HTML 4 as an XML 1.0 application
- → XHTML 1.0, 1.1 (2.0)
- Stricter syntax – XML rules

- **Outdated approach**

XML Syntax

XML

- Based on **tags**, **elements**, **attributes**, entities and text
- Basis – no tags predefinition
- XML prolog (optional | “should include”)
  - <?xml version="1.0" encoding="utf-8"?>
- XML Does Not DO Anything

XML Syntax - characters

Parentheses	( )
Square Brackets	**[ ]**
Angle Brackets	**< >**
Braces	{ }
Slashes	/ \
Quotes	" (not „“)

Characters II

Comma	,
Single quotes, apostrophe	', ‘
Dash, underscore	-, _
Colon	:
Semicolon	;
Equal sign	=
Exclamation/question mark	!/?

XML Syntax: **Prolog**

```
<?xml version="1.0" encoding="UTF-8"?>
<root_element>
```

XML Syntax: **Element**

Element

Start tag

Content

End tag

```
<p class="foo">This is a paragraph.</p>
```

Attribute name

value

Element

opening tag closing tag

<name>...</name>

element name | content | forward slash

Attribute and value

```
<elm some_name="value"
```

attribute name

quotes

Syntax – Elements

- Element always have closing tags
  - <tag>This is paragraph. **</tag>**
  - **<tag attributes="..”/>**
- Proper nesting (no overlapping)
  - **Wrong **
  - **Right**
- All tags are case sensitive
  - <Library>Tag
  - <library>Different tag

Tag Names

- Allowed characters
  - Letters, Numbers,Other characters
- At the beginning of tag
  - Only letter
  - No xml
    - **wrong <xmltext>, right <textxml>**
- No spaces
- Recommendation - Small letters, No national characters, only **a-z**

Recommendations for Naming Tags

- Descriptive name
  - <date_of_birth> vs <date>
- Short and brief
  - <name>
  - <name_of_my_old_friend_from_childhood>
- Problems
  - <book-name> - some programs can separate
  - <book.name> - may be wrong
  - <book:name> - namespaces

Syntax - Attributes

- Attribute values are in quotes
  - **<img src=image.jpg />** wrong
  - **<img src="image.jpg" />** right
  - double **(" ")** or single **(' ')** quotes

Usage of Attributes

- Adding item into group or class
- Identification of item
- Reference to an item
- Data frequently used by searching
- Complementary information
- Depends on author and an application

Advantages of Elements in Comparison to Attributes

- Attributes are limited in multiple values
- Attribute does not cover data structure
- Attribute is **not extendable**
- Attribute can be more difficult to read

```
<note day="10" month="01" year="2008"
to="Tove" from="Jani" heading="Reminder"
body="Don't forget me this weekend!">
</note>
```

XML Syntax: **Comments**

```
<?xml version="1.0" encoding="UTF-8"?>
<root_element>
      <child>
            <subchild>.....</subchild>
            <subchild>.....</subchild>
      </child>
      <!– this comment is ignored -->
</root_element>
```

Comments

- Comments may appear anywhere in a document.
- Exceptions
  - Before XML declaration.
  - <Within tags>
  - Within attribute values.
  - Nested inside the other comments.

Multiline comments

```
<?xml version="1.0" encoding="UTF-8"?>
<root_element>
        <child>…        </child>
        <!-
                this is the
                multi
                line
                comment
        -->
</root_element>
```

Ca we use quotes” within an attribute value?

Entities

- Special symbols for adding predefined content
- For reserved symbols of XML
- Some predefined
- Custom entities

Fixed Entities

- < less than - `<`
- & ampersand - `&`
- > greater than - `>`
- “ double quote - `"`
- ` single quote - `'`

CDATA Sections

- Character data
- Parts not processed by a parser
- For example to include HTML code

```
<my_html>
        <![CDATA[
                <p>Some text…</p>
        ]] >
</my_html>
```

Android strings.xml Example

```
    </string>
    <string name="czu">ČZU</string>
    <string name="dejvicka">Dejvická</string>
    <string name="busz">107 z</string>
    <string name="onas">O nás</string>
    <string name="feedback">Váš názor</string>
    <string name="domu">Domů</string>

    <string-array name="pruvodce_obsah">
        <item>
<![CDATA[
<p>Během studia se musíte řídit <a href=\"http://dl.webcore.czu.cz/file/bmZCQ3ozYVlHanM9\">Studijním a
<p> </p>
<p>V rámci svého studia, pokud studujete první vysokou školu, máte nárok na:</p>
<ul>
<li>3 + 1 rok bezplatného studia v bakalářském programu</li>
<li>2 + 1 rok bezplatného studia v magisterském programu</li>
</ul>
<p> </p>
<p>Pro postup do vyššího ročníku musíte splnit úkoly předepsané ve studijním <br />plánu. V zimním sem
<p> </p>
<p>Před zápisem do vyššího ročníku musíte předložit výkaz o studiu se splněným studijním plánem na stu
<p> </p>
<p>Pokud nesplníte podmínky postupu do vyššího ročníku, můžete požádat o opakování ročníku.</p>
]]>
        </item>
```

Document Structure

```
<?xml version="1.0" encoding="UTF-8"?>
<root_element>
    <child>
        <subchild>.....</subchild>
        <subchild>.....</subchild>
    </child>
</root_element>
```

Root Element of XML

Sub-Element of <student> Node

Attribute

```
<student>
    <male std="1st">
      <name>John</name>
      <age>6</age>
      <marks>90.45</marks>
    </male>
</student>
```

Sub-Element of <male> Element

www.c4learn.com

www.c4learn.com

Text Value / Number Value of <marks> element

Is the XML document OK?

Validation, well formedness

Correct Document is

Well formed

- **XML syntax**
- Basic XML rules and principles
- One main element

Valid

- Validation against some prescription
- Structure of some specific document
- DTD
- **XML Schema**

XML Namespaces

<prefix:element_name />

xml:namespaces

- When combining XML documents together
- Name conflicts

  → use namespaces

employeeTable

empID	secID	name
E0000001	S001	John Smith
E0000002	S002	Ichiro Tanaka
E0000003	S002	Jiro Suzuki
E0000004	S003	Saburo Takahashi

sectionTable

secID	name
S001	Sales
S002	Development
S003	Administrative

```
SELECT employee.empID,section.name,employee.name
FROM employee,section
WHERE employee.secID = section.secID
```

```
SELECT e.empID,s.name,e.name
FROM employee e,section s
WHERE e.secID = s.secID
```

empID	s.name	e.name
E0000001	Sales	John Smith
E0000002	Development	Ichiro Tanaka
E0000003	Development	Jiro Suzuki
E0000004	Administrative	Saburo Takahashi

XML Namespaces

<element xmlns:**prefix**="URI" >

<**prefix**:elementName ….

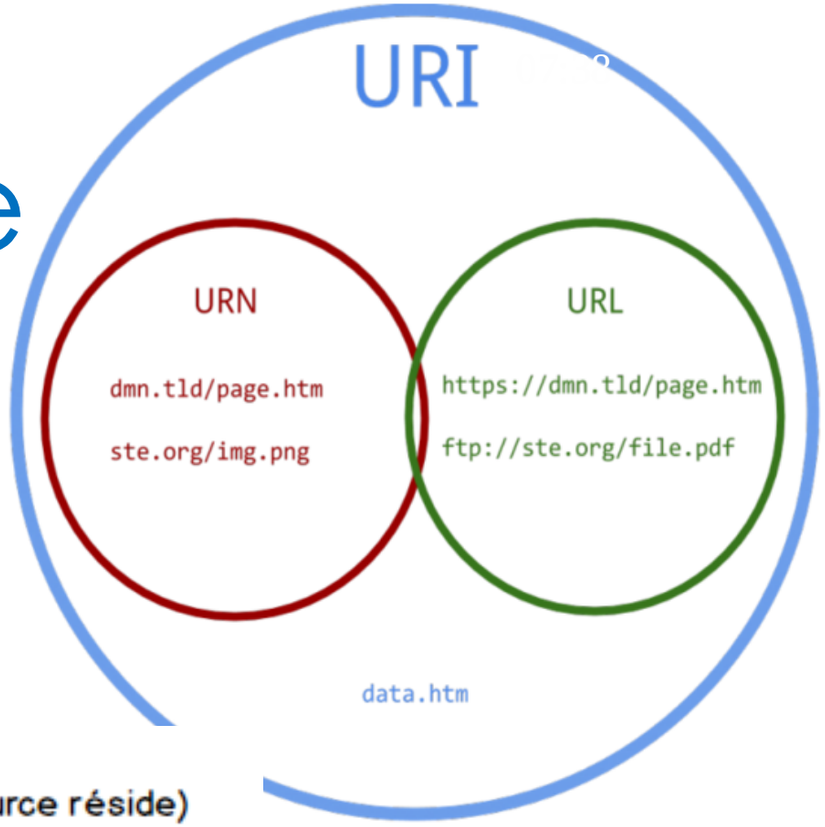
- All child elements with the same prefix are associated with the same namespace
- Real use – XML Schema

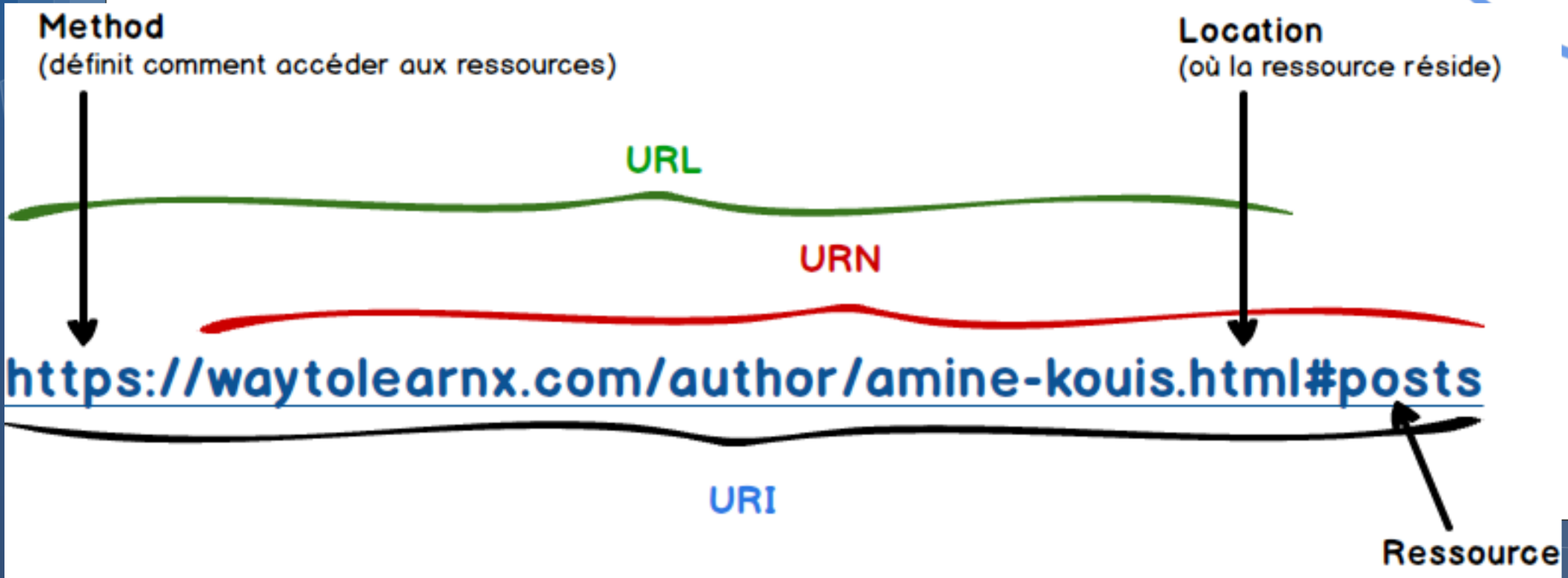
<element xmlns:prefix="URI" >

- xmlns
  - Definition of a namespace
  - “xmlns” does not change
- prefix
  - Some prefix to be used (prefix:, h:, tab:, xsd:)
- URI
  - Identifier of a namespace
  - Not used to look up by parser

URI – Unified Resource Identifier

- String which unambiguously identifies resource





Name Conflicts

```
<table>
 <tr>
   <td>Apples</td>
   <td>Bananas</td>
 </tr>
</table>
```

```
<table>
  <name>
        African Coffee
        Table
   </name>
  <cell>80</cell>
  <cell>120</cell>
</table>
```

Solution

```
<root>
    <h:table xmlns:h="http://www.w3.org/TR/html4/">
      <h:tr>
        ........
      </h:tr>
    </h:table>
    <f:table xmlns:f="http://www.w3schools.com/furniture">
      <f:name>African Coffee Table</f:name>
      ........
    </f:table>
</root>
```

Namespace Can be Defined in Root Element

```
<root
xmlns:h="http://www.w3.org/TR/html4/"
xmlns:f="http://www.w3schools.com/furniture">

<h:table>
  <h:tr>
  …
```

Default Namespace

```
<table
xmlns="http://www.w3.org/TR/html4/">
  <tr>
    <td>Apples</td>
    <td>Bananas</td>
  </tr>
</table>
```

DTD

Document Type Definition

DTD – Document Type Definition

- Special language to define document structure
- Flexible tool
- Definition of tags
- Definition of attributes
- Definition of entities

DTD is for XML **validation!**

DTD Validation – Where?

- By an application
- Inside certain editors
  - (PHP Storm, NetBeans, XML Copy Editor, etc.)
- Online services
  - http://www.xmlvalidation.com
  - http://www.validome.org/grammar/validate/
  - https://validator.w3.org/

DTD Defines

- Elements
  - Position
  - Children
  - Content (limited)
- Attributes
  - Content (limited)
- Entities

DTD Defines

- Also unused elements or attributes (if permitted)
- What is not permitted is prohibited
- All parts of XML must agree to the DTD
- When processing error - break

DTD Is Not Able to Define

- Exact number of elements (easily)
- What will be the result
- What is the meaning of values
- The content (content types)
- The allowed values

Where to Place the DTD

- Beginning of document
- External file
- Combination

Beginning of the Document

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE person [
    <!ELEMENT person (name, profession*)>
    <!ELEMENT name (firstname, surname)>
    <!ELEMENT firstname (#PCDATA)>
    …
]>
<person>
……
```

Declaration of DTD Externally

<?xml version="1.0"encoding="UTF-8"?>

<!DOCTYPE person SYSTEM *"path"*>

- Immediately after xml declaration
- SYSTEM – non public standard
- PUBLIC – for public standard – (SVG, RSS, …)

Combination

```
<!DOCTYPE person (SYSTEM „link.dtd“)[
   <!ELEMENT person (name, profession*)>
   <!ELEMENT name (firstname, surname)>
   …
]>
<person>
….
```

Elements

Declaration

- Syntax: **<!ELEMENT element_name (content)>**
- Content:
- #PCDATA/#CDATA – already indivisible
  - <!ELEMENT phone (#PCDATA)>
  - <phone>my graet cell phone</phone>
- Another elements
  - <!ELEMENT person (firstname, surname)>
  - <person><firstname/><surname/></person>

Declaration of Element

- Selection
  - <!ELEMENT month (janury | february | ...)>
- Combination
  - <!ELEMENT circle (center, (diameter | radius))>
- Mixed content
  - <!ELEMENT definition (#PCDATA | term)*>
- Empty element
  - <!ELEMENT image EMPTY>
- General element
  - <!ELEMENT page ANY>

The Type of Element Content

- PCDATA (parsed character)
  - Processed by parser
  - Text with entities
  - Without nested elements
  - <!ELEMENT phone (#PCDATA)>
- CDATA (character data)
  - Data are not processed parser
  - HTML code
  - JavaScript
  - …

Element Qualifiers - Number of Occurrences

* - none or more

+ - one or more

? – none or just one

- If not listed - exactly one

<!ELEMENT university(building+,address,student*,brewery?)>

What´s Wrong?

(building+, address, student*, brewery?)

```
<university>
      <address> </address>
      <student></student>
</university>
```

What´s Wrong?

(building+,address,student*,brewery?)

```
<university>
      <building></building>
      <address> </address>
      <address> </address>
      <student></student>
      <brewery></brewery>
</university>
```

What´s Wrong?

(building+,address,student*,brewery?)

```
<university>
      <building></building>
      <address> </address>
      <student></student>
      <brewery></brewery>
      <brewery></brewery>
</university>
```

What´s Wrong?

(building+,address,student*,brewery?)

```
<university>
      <brewery></brewery>
     <building></building>
     <address> </address>
     <student></student>
</university>
```

What´s Wrong?

(building+,address,student*,brewery?)

```
<university>
      <building></building>
      <address> </address>
      <student></student>
      <student></student>
      <brewery></brewery>
</university>
```

Attributes

Declaration of Attributes

```
<!ATTLIST tag_name attribute_name
    type neccesity default_value >
```

- One declaration for more attributes

```
<!ATTLIST img src CDATA #REQUIRED
                  width CDATA #REQUIRED
                  height CDATA #REQUIRED
>
```

Content Types

- <> data types
- CDATA – general text
- Enumeration
- ID
- IDREF, IDREFS
- NMTOKEN, NMTOKENS
- ENTITY, NOTATION

<!ATTLIST tag_name attribute_name type neccesity def_value >

Enumeration

<!ATTLIST date month (january | february | …)

- List of possible values

ID

- A unique value within the document
  - Regarding the attribute
- Similar to the primary key
- For data that we want to distinguish
- For example, database of students
- <!ATTLIST student ID #REQUIRED>

IDREF

- Similar to a foreign key in the database
- It must contain the ID value of another attribute

<project number=„C3“>ID
<name>strategic planning</name>
<member code=„k28“/></project> IDREF

<scientist code=„k28“>ID
<name>John Smith</name>
<assigned project=„C3“/> IDREF
</scientist>

IDREFS

- Same like IDREF
- It can have more values

NMTOKEN

- A text string
- The value is a valid XML name
- Limitation
  - Just text, number, underscore
- NMTOKEN can‘t contain
  - Space
  - Other special characters
- <name>John</name>
- <something attr=„name“></….

NMTOKENS

- Same like NMTOKEN
- Contains more values
- Values are separated with space

Necessity of Attributes

- #REQUIRED – required
- #IMPLIED – optional
- #FIXED + default value
  - unchangeable - the reference to standard

```
<!ATTLIST tag_name attr_name type necessity
def_value >
```

Entities

Declaration of Entity

- `<!ENTITY entity_name „your special value“>`
- `&entity_name;`
- You can insert a long string, such as HTML footer of the page

Fixed Entities

- < less than - <
- & ampersand - &
- > greater than - >
- “ double quote - "
- ` single quote - '

External Entity

- Paste the contents of another file
- <!ENTITY footer SYSTEM „http://something.com/footer.xml“>
- More about entities
- http://www.tutorialspoint.com/dtd/dtd_entities.htm

Questions?